

★ WEEK 14 · PART 1 · PRACTICE

Sample, Don't Integrate - Worked by Hand

Answers as Averages of Draws, and How Many You Need

Monte Carlo estimation and the $1/\sqrt{N}$ law

Part 1's core move, with the arithmetic shown. We turn a posterior integral into an average of draws - reading off the mean and the probability of profit - then work out how many samples precision demands, and why correlated MCMC draws cost more.

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PART 1 · APPLICATIONS

Turn Integrals Into Averages

Case A: Monte Carlo estimation - the posterior mean and the probability of profit, each just a summary of draws.
Case B: convergence - the $1/\sqrt{N}$ law, the samples a target precision needs, and the price of correlated draws.
Then a Claude Code lab.

What Part 1 Told Us

Now we make it concrete - with numbers

1

An integral becomes an average

Monte Carlo replaces a hard posterior integral with an easy average over draws. That single move unblocks everything.

2

Every answer is a summary of draws

The mean, an interval, the probability of profit - all are just summaries of the same set of posterior samples.

3

Draws can be correlated but still valid

MCMC draws are dependent, so they carry less information each. You need more of them for the same precision.

Everything here is a calculator exercise: a handful of draws, a fraction, and a sample count.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a set of posterior draws.
- Summarize them, or count how many you need.
- Then read the answer or the cost.

What you should watch

- How an expectation becomes an average.
- How a probability becomes a fraction.
- How the error falls as 1 over root- N .

Answers From Draws

The mean, and the probability of profit

Once you have posterior draws of a strategy's expected return, every question is a summary of them. The posterior mean is their average. The probability the strategy is profitable is the fraction of draws above zero. And the Monte Carlo error tells you how much to trust that fraction. We compute all three by hand

The draws

- A small set, to show the averaging.
- Then a full set of 1000 draws.
- Of the expected annual return.

What we'll do by hand

- Average the draws for the mean.
- Count the winners for $P(\text{profitable})$.
- Compute the Monte Carlo error.

Monte Carlo

Integrals become averages

$$\mathbb{E}[f(\theta)] = \int f(\theta) p(\theta | D) d\theta \approx \frac{1}{N} \sum_{i=1}^N f(\theta^{(i)}) \quad (\text{a hard integral becomes an easy average})$$

The move

- The expectation is an integral over the posterior.
- Replace it with an average over draws.
- The more draws, the better the average.

Why it works

- The sample average estimates the true mean.
- It converges as the draw count grows.
- Hard analysis becomes easy arithmetic.

The Posterior Mean

Just average the draws

$$\hat{\mathbb{E}}[\text{return}] = \frac{5+8+11+6+10}{5} = 8.0\% \quad (\text{the posterior mean is just the sample average})$$

The calculation

- Five draws: 5, 8, 11, 6, and 10 percent.
- Their sum is 40.
- Divided by five, the mean is 8%.

The point

- No integral was solved.
- The mean is a plain average.
- The same works for thousands of draws.

The Probability of Profit

Count the winning draws

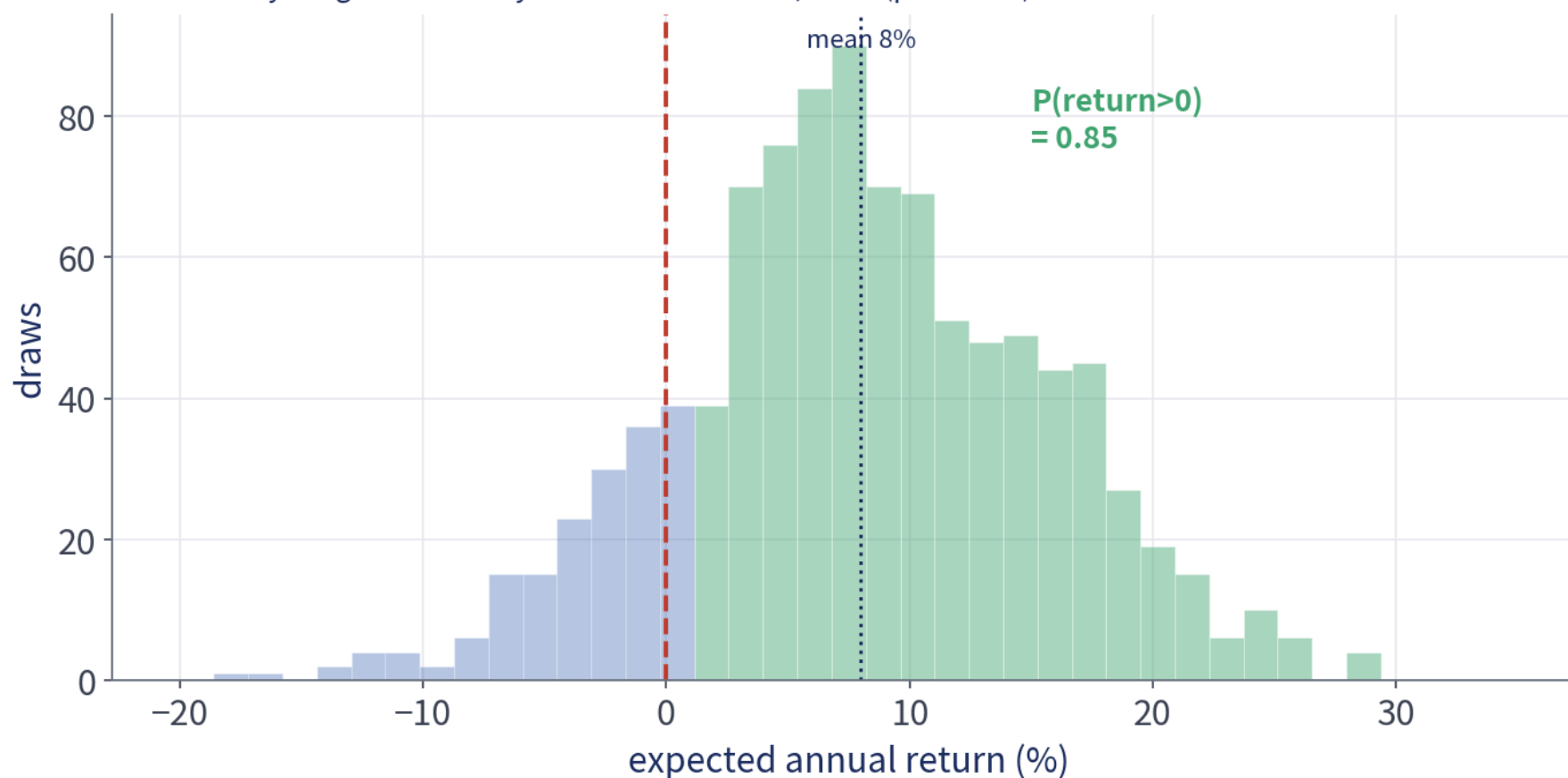
$$P(\text{return} > 0) = \frac{850}{1000} = \mathbf{0.85} \quad (\text{just count the winning draws})$$
$$\text{MC error} = \sqrt{\frac{0.85(0.15)}{1000}} = \mathbf{1.1\%}$$

Of 1000 draws, 850 are positive, so the probability the strategy is profitable is 0.85 - with a Monte Carlo error of about 1.1%. A probability is just a fraction of draws.

Everything Is a Summary of Draws

The histogram of posterior draws is the whole answer. Its balance point is the mean (8%); the green fraction to the right of zero is the probability of profit (0.85); and the spread tells you the uncertainty. One set of draws answers every question you can ask

Case A: everything is a summary of draws - the mean, and P(profitable)



Sample, Then Summarize

The whole plan in four words

What this buys you

- No integrals to solve, ever.
- Any quantity is a summary of the draws.
- Means, intervals, probabilities - all the same set.

Why it matters

- A profit probability is a decision input.
- It comes free from the draws you already have.
- The hard part is getting good draws.

How Many Draws?

Precision, and the cost of correlation

Monte Carlo answers come with an error that shrinks as 1 over the square root of the sample count - so quadrupling the draws only halves the error. We work out how many independent draws a target precision needs, then see why correlated MCMC draws, carrying less information each, demand far more

The setup

- Per-draw posterior spread of 4%.
- A target standard error of 0.1%.
- MCMC draws with 0.8 autocorrelation.

What we'll do by hand

- Apply the 1-over-root-N law.
- Solve for the samples needed.
- Adjust for the effective sample size.

The 1-Over-Root-N Law

Error and sample count

$$SE = \frac{\sigma}{\sqrt{N}} \Rightarrow \text{quadruple } N \text{ to halve the error}$$

The scaling

- The error is the spread over root-N.
- It falls slowly - only as a square root.
- Quadruple the draws to halve the error.

The implication

- High precision is expensive.
- Each extra digit costs a hundredfold.
- So efficient sampling matters.

Samples for a Target

Solve the law for N

$$N = \left(\frac{\sigma}{\text{SE}} \right)^2 = \left(\frac{4\%}{0.1\%} \right)^2 = 40^2 = \mathbf{1600} \text{ independent samples}$$

To pin the standard error to 0.1% with a 4% per-draw spread, you need 1600 independent draws - the ratio of spread to target, squared.

The Effective Sample Size

Correlated draws count for less

$$\text{ESS} = N \frac{1-\rho}{1+\rho} = N \frac{0.2}{1.8} = \frac{N}{9} \Rightarrow 1000 \text{ MCMC draws} \approx 111 \text{ independent}$$

The correction

- Autocorrelation shrinks the effective count.
- At 0.8, the factor is one-ninth.
- So 1000 draws act like about 111.

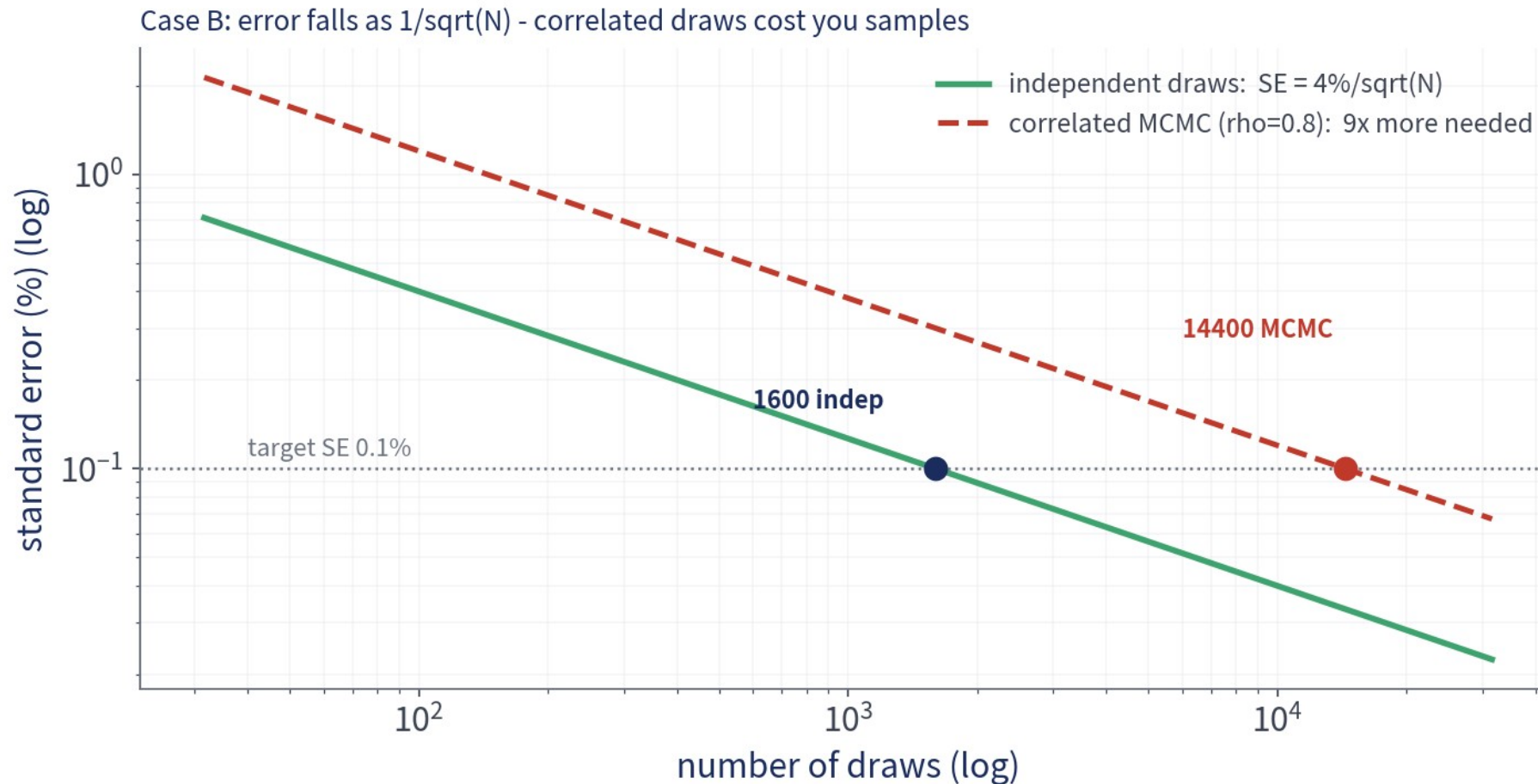
The cost

- You need nine times as many MCMC draws.
- That's 14,400 for the same precision.
- Correlation is the tax MCMC pays.

Error Falls Slowly, Correlation Costs

On a log-log plot the standard error is a straight line falling as $1/\sqrt{N}$. Independent draws hit the 0.1% target at 1600; correlated MCMC draws, each worth a ninth, need 14,400 to reach the same line.

The gap is the price of dependence



Correlation Is a Tax

Valid, but expensive

Why it matters

- MCMC draws are dependent by construction.
- Each carries less new information.
- So the effective sample size is smaller.

What to do

- Report effective sample size, not raw count.
- Reduce autocorrelation with better samplers.
- That's exactly what HMC and NUTS aim for.

Sample, Don't Integrate

The whole strategy in one line

Both cases rest on one idea: never solve the integral, sample it.

Draws give you every answer as a summary - a mean, a probability, an interval. The only question is how many draws you need, which the 1-over-root-N law answers - inflated by any correlation among them. Get good draws and the rest is arithmetic.

Estimation (Case A)

- The mean is an average of draws.
- A probability is a fraction of draws.
- The error is the spread over root-N.

Convergence (Case B)

- Error falls as 1 over root-N.
- A target precision fixes the count.
- Correlation inflates what you need.

Reproduce the Estimates and the Counts

Direct Claude Code to check your arithmetic

GOAL Verify the Monte Carlo estimates and the sample-count arithmetic, then simulate.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - estimation:

- * average the draws [5,8,11,6,10] for the mean;
- * from 1000 draws with 850 positive, print $P(\text{profit})$ and its MC standard error.

CASE B - convergence:

- * with per-draw sd 4%, print SE for $N=100, 400, 1000$;
- * solve for N at $SE=0.1\%$;
- * with $\rho=0.8$, print the ESS factor and the MCMC draws needed.

Then draw samples and confirm the $1/\sqrt{N}$ law.

Check against the slides

- mean 8%, $P(\text{profit})$ 0.85.
- MC error $\sim 1.1\%$.
- $N = 1600$ independent.
- $ESS = N/9 \rightarrow 14,400$ MCMC.

Deliverable: `part1_practice.py` whose printed numbers match the estimates and the sample counts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How does an expectation become an average?

**Monte Carlo: average
f over the draws**

Q2 How does the error scale with sample count?

**As 1 over the square
root of N**

Q3 Why do correlated draws cost more?

**The effective sample
size is smaller**

All three come from Monte Carlo estimation and the 1-over-root-N law.

Three Pitfalls to Avoid

Where sampling trips people

Common mistakes

- "More draws always fix the error fast."
- "Report a Monte Carlo number with no error."
- "Count MCMC draws as if independent."

The reality

- Error falls only as root-N - it's slow.
- Always attach the Monte Carlo standard error.
- Use effective sample size for correlated draws.

Two Calculations to Remember

Part 1 applications in one screen

1

Every answer is a summary of draws

The posterior mean is the average of the draws (8%), the probability of profit is the fraction above zero (0.85), and its Monte Carlo error is about 1.1%. No integral is ever solved - the samples carry everything.

2

Precision costs samples, correlation costs more

Error falls only as $1/\sqrt{N}$, so quadrupling draws halves the error; pinning it to 0.1% needs 1600 independent draws. With 0.8 autocorrelation the effective sample size is a ninth, so MCMC needs 14,400.

3

Sample, don't integrate

The strategy is to replace an intractable integral with an average over draws. The only real question is how to generate good, low-correlation draws - which is what the rest of the week builds.

Next — Part 2 practice: the Metropolis and Gibbs samplers that actually generate the draws.